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Brightlearn
Exercise number 4
JOINS

Part A: INNER JOIN

Question 01: Show every user who has a subscription. Match users to subscriptions.

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78
79 ---- part A, INNER JOIN
80 --- question 1: show every user who has a subscription. match users to subscriptions.
81 ✓ SELECT users.user_id,
82         users.user_name,
83         subscriptions.subscription_id,
84         subscriptions.start_date
85 FROM exercise_n04.joins.users AS users
86 INNER JOIN exercise_n04.joins.subscriptions AS subscriptions
87 ON users.user_id = subscriptions.user_id;
```

Add parameter

Table ▾ +

	¹ ₃ user_id	^A _B user_name	¹ ₃ subscription_id	 start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	1	Nomvula	501	2026-01-15

Question 02: Show every subscription with its matching plan name and monthly price.

```
87 ON users.user_id = subscriptions.user_id;
88 --- question 2: show every subscription with its matching plan name and monthly price.
89 SELECT subscriptions.subscription_id,
90        subscriptions.user_id,
91        plans.plan_name,
92        plans.monthly_price
93 FROM exercise_n04.joins.subscriptions AS subscriptions
94 INNER JOIN exercise_n04.joins.plans AS plans
95 ON subscriptions.plan_id = plans.plan_id;
```

Add parameter

Table	+			
	1 ² subscription_id	1 ² user_id	A ^B plan_name	1 ² monthly_price
1	501	1	Basic	79
2	504	6	Standard	129
3	503	1	Premium	199
4	505	3	Family	249
5	502	2	Standard	129

Question 03: Show every viewing session that has a matching show. Include the title and genre.

```

96  --- question 3: show every viewing session that has a matching show. include the show title and genre
97  SELECT viewing_sessions.session_id,
98         viewing_sessions.user_id,
99         shows.show_title,
100        shows.genre,
101        viewing_sessions.watch_minutes
102  FROM exercise_n04.joins.viewing_sessions AS viewing_sessions
103  INNER JOIN exercise_n04.joins.shows AS shows
104  ON viewing_sessions.show_id = shows.show_id;
105  |

```

Add parameter

	1 ² session_id	1 ² user_id	A ^B show_title	A ^B genre	1 ² watch_minutes
1	904	7	Comedy Hour	Comedy	20
2	903	1	Crime Time	Drama	60
3	902	2	Tech Tales	Documentary	30
4	901	1	Comedy Hour	Comedy	45

Question 04: Show viewing session with the user who watched it. Only show sessions with a matching user.

```

105  --- question 4: show every viewing session with the user who watched it. only show sessions with a matching user
106  SELECT users.user_name,
107         users.country,
108         viewing_sessions.session_id,
109         viewing_sessions.watch_minutes
110  FROM exercise_n04.joins.users AS users
111  INNER JOIN exercise_n04.joins.viewing_sessions AS viewing_sessions
112  ON users.user_id = viewing_sessions.user_id;
113  |

```

Add parameter

	A ^B user_name	A ^B country	1 ² session_id	1 ² watch_minutes
1	Nomvula	Johannesburg	903	60
2	David	Cape Town	902	30
3	Anele	Durban	905	90
4	Nomvula	Johannesburg	901	45

Question 05: Show users along with their subscriptions, the plan name, and price. Use only users who have both a subscription and a valid plan.

```

113 --- question 05: show users along with their subscriptions, the plan name, and price. use only users who have both a
    subscription and a valid plan.
114 SELECT users.user_name,
115         users.country,
116         plans.plan_name,
117         plans.monthly_price,
118         subscriptions.start_date
119 FROM exercise_n04.joins.users AS users
120 INNER JOIN exercise_n04.joins.subscriptions AS subscriptions
121 ON users.user_id = subscriptions.user_id
122 INNER JOIN exercise_n04.joins.plans AS plans
123 ON subscriptions.plan_id = plans.plan_id;

```

Add parameter

	user_name	country	plan_name	monthly_price	start_date
1	Nomvula	Johannesburg	Premium	199	2026-03-10
2	David	Cape Town	Standard	129	2026-02-01
3	Anele	Durban	Family	249	2026-04-05
4	Nomvula	Johannesburg	Basic	79	2026-01-15

Part B: LEFT JOIN

Question 06: Show every user and any subscription they have. Users without subscription must still appear.

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124 --- question 6: show every user and any subscription they have. users without subscriptions must still appear
125 SELECT users.user_id,
126         users.country,
127         subscriptions.subscription_id,
128         subscriptions.start_date
129 FROM exercise_n04.joins.users AS users
130 LEFT JOIN exercise_n04.joins.subscriptions AS subscriptions
131 ON users.user_id = subscriptions.user_id;
132
133

```

Add parameter

	user_id	country	subscription_id	start_date
1	1	Johannesburg	503	2026-03-10
2	2	Cape Town	502	2026-02-01
3	3	Durban	505	2026-04-05
4	4	Pretoria	null	null
5	5	Port Elizabeth	null	null
6	1	Johannesburg	501	2026-01-15

Question 07: Show every plan and the subscription on it. Plans with no subscriptions must still appear.

```

132 --- question 7: Show every plan and the subscription on it. Plans with no subscriptions must still appear.
133 SELECT plans.plan_id,
134        plans.plan_name,
135        subscriptions.subscription_id,
136        subscriptions.user_id
137 FROM exercise_n04.joins.plans AS plans
138 LEFT JOIN exercise_n04.joins.subscriptions AS subscriptions
139 ON plans.plan_id = subscriptions.plan_id;
140

```

Add parameter

	plan_id	plan_name	subscription_id	user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

Question 08: Show every show and any viewing sessions on it. Shows that were never watched must still appear.

```

140 --- question 8: Show every show and any viewing sessions on it. Shows that were never watched must still appear.
141 SELECT shows.show_id,
142        shows.show_title,
143        viewing_sessions.session_id,
144        viewing_sessions.watch_minutes
145 FROM exercise_n04.joins.shows AS shows
146 LEFT JOIN exercise_n04.joins.viewing_sessions AS viewing_sessions
147 ON shows.show_id = viewing_sessions.show_id;
148

```

Add parameter

	show_id	show_title	session_id	watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45

Question 09: Show every viewing session and the user who watched it. Sessions referencing users that do not exist must still appear (with NULL user details).

```

148
149 --- question 09: Show every viewing session and the user who watched it. Sessions referencing users that do not exist must
    still appear (with NULL user details).
150 SELECT viewing_sessions.session_id,
151        viewing_sessions.watch_minutes,
152        users.user_id,
153        users.user_name
154 FROM exercise_n04.joins.viewing_sessions AS viewing_sessions
155 LEFT JOIN exercise_n04.joins.users AS users
156 ON viewing_sessions.user_id = users.user_id;

```

Add parameter

	1 ² session_id	1 ² watch_minutes	1 ² user_id	1 ⁸ user_name
1	901	45	1	Nomvula
2	902	30	2	David
3	903	60	1	Nomvula
4	904	20	null	null
5	905	90	3	Anele

Question 10: Show every user, the plan they are on (if any) , and the monthly prices. Users without a subscription must still appear.

```

157 ---question 10: Show every user, the plan they are on (if any) , and the monthly prices. Users without a subscription must
    still appear.
158 SELECT users.user_id,
159        users.country,
160        plans.plan_name,
161        plans.monthly_price
162 FROM exercise_n04.joins.users AS users
163 LEFT JOIN exercise_n04.joins.subscriptions AS subscriptions
164 ON users.user_id = subscriptions.user_id
165 LEFT JOIN exercise_n04.joins.plans AS plans
166 ON subscriptions.plan_id = plans.plan_id;

```

Add parameter

	1 ² user_id	1 ⁸ country	1 ⁸ plan_name	1 ² monthly_price
1	1	Johannesburg	Premium	199
2	2	Cape Town	Standard	129
3	3	Durban	Family	249
4	4	Pretoria	null	null
5	5	Port Elizabeth	null	null
6	1	Johannesburg	Basic	79

Part c: FULL OUTER JOIN

Question 11: Show every user and every subscription, including users without subscriptions and subscriptions referencing users that do not exist.

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166 ON subscriptions.plan_id = plans.plan_id;
167 --- QUESTION 11: Show every user and every subscription, including users without subscriptions
    and subscriptions referencing users that do not exist.
168 SELECT users.user_id,
169        users.user_name,
170        subscriptions.subscription_id,
171        subscriptions.start_date
172 FROM exercise_n04.joins.users AS users
173 FULL OUTER JOIN exercise_n04.joins.subscriptions AS subscriptions
174 ON users.user_id = subscriptions.user_id;

```

	1 ² user_id	1 ² user_name	1 ² subscription_id	1 ² start_date
1	5	Lerato	null	null
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	1	Nomvula	501	2026-01-15
7	null	null	504	2026-03-20

Question 12: Show every plan and every subscription, including plans without subscribers AND any subscription referencing a plan that does not exist.

```

175 --- question 13: Show every plan and every subscription, including plans without subscribers AND any subscription referencing a plan that does not exist.
176 ✓ SELECT plans.plan_id,
177        plans.plan_name,
178        subscriptions.subscription_id,
179        subscriptions.user_id
180 FROM exercise_n04.joins.plans AS plans
181 FULL OUTER JOIN exercise_n04.joins.subscriptions AS subscriptions
182 ON plans.plan_id = subscriptions.plan_id;
183

```

Add parameter

	1 ² plan_id	1 ² plan_name	1 ² subscription_id	1 ² user_id
1	12	Premium	503	1
2	10	Basic	501	1
3	13	Family	505	3
4	14	Mobile	null	null
5	11	Standard	504	6
6	11	Standard	502	2

Question 13: Show every show and viewing session, including shows that were never watched AND sessions referencing shows that do not exist

```

--- question 13: Show every show and viewing session, including shows that were never watched AND sessions referencing shows that do not exist
✓ SELECT shows.show_id,
       shows.show_title,
       viewing_sessions.session_id,
       viewing_sessions.watch_minutes
FROM exercise_n04.joins.shows AS shows
FULL OUTER JOIN exercise_n04.joins.viewing_sessions AS viewing_sessions
ON shows.show_id = viewing_sessions.show_id;

```

	1 ² show_id	1 ² show_title	1 ² session_id	1 ² watch_minutes
1	704	Cooking Lab	null	null
2	703	Tech Tales	902	30
3	701	Comedy Hour	904	20
4	702	Crime Time	903	60
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45
7	null	null	905	90

Question 14: Show every user and viewing session, including users with no sessions AND sessions referencing users who do not exist.

```

1  --- question 14: Show every user and viewing session, including users with no sessions AND sessions referencing users who do not exist
2  SELECT users.user_id,
3         users.user_name,
4         viewing_sessions.session_id,
5         viewing_sessions.show_id,
6         viewing_sessions.watch_minutes
7  FROM exercise_n04.joins.users AS users
8  FULL OUTER JOIN exercise_n04.joins.viewing_sessions AS viewing_sessions
9  ON users.user_id = viewing_sessions.user_id;
10

```

	user_id	user_name	session_id	show_id	watch_minutes
1	1	Nomvula	901	701	45
2	1	Nomvula	903	702	60
3	3	Anele	905	705	90
4	2	David	902	703	30
5	null	null	904	701	20
6	4	Kabelo	null	null	null
7	5	Lerato	null	null	null

Question 15: Show every user, every subscription, and every plan in one query- using FULL OUTER JOIN throughout.

```

200  --- question 15: Show every user, every subscription, and every plan in one query- using FULL OUTER JOIN throughout.
201  SELECT users.user_id,
202         users.user_name,
203         subscriptions.subscription_id,
204         plans.plan_id,
205         plans.plan_name
206  FROM exercise_n04.joins.users AS users
207  FULL OUTER JOIN exercise_n04.joins.subscriptions AS subscriptions
208  ON users.user_id = subscriptions.user_id
209  FULL OUTER JOIN exercise_n04.joins.plans AS plans
210  ON subscriptions.plan_id = plans.plan_id;
211

```

	user_id	user_name	subscription_id	plan_id	plan_name
1	1	Nomvula	503	12	Premium
2	1	Nomvula	501	10	Basic
3	3	Anele	505	13	Family
4	5	Lerato	null	null	null
5	4	Kabelo	null	null	null
6	2	David	502	11	Standard
7	null	null	504	11	Standard
8	null	null	null	14	Mobile

Bonus Challenge

Bonus 01: Which users have not subscribed to any plan?

user_id	user_name
4	Kabelo
5	Lerato

Bonus 02: Which subscriptions reference users that do not exist in the user table?

Subscription_id	User_id
504	6

Bonus 03: Which shows have never been watched?

Show_id	show_title
706	Wild earth

Bonus 04: Which viewing sessions referencing shows that do not exist?

Show_id	Session_id
705	905

Bonus 05: Which plans have no subscribers?

Plan_id	Plan_name
14	Mobile